Mathematics for Aerospace Engineers

Chapter 4: Multiple integration

School of Mathematical Sciences

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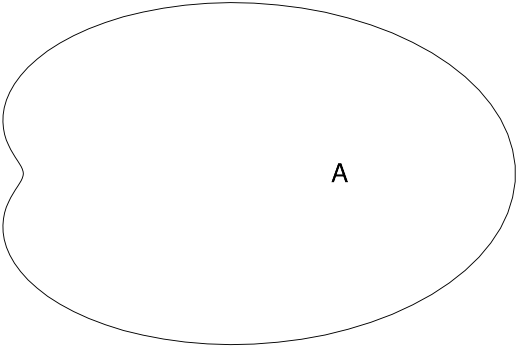
Overview

4 Multiple Integration

4.1 Introduction

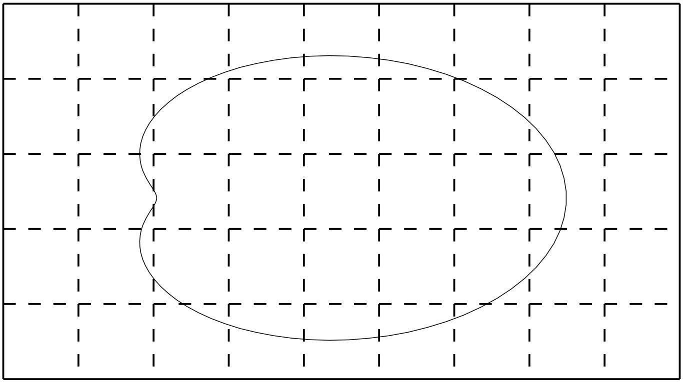
Consider the following problem:

What is the mass M of a thin plate (called a lamina) with variable density $\rho(x, y)$ occupying the region A?



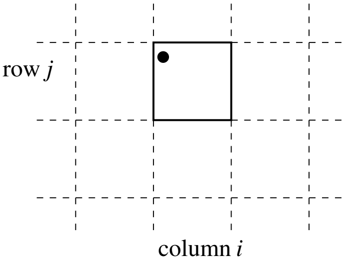
Step 1:

- Divide up the region A into small subregions using a *grid* or *mesh*.



Step 2:

- Consider the rectangle in the ith column and the jth row. Let the area of this rectangle be ΔA_{ij}.
- Take any point (x_i, y_j) in this rectangle
- Assume the density is constant throughout the rectangle
- Assume this constant density is $\rho(x_i, y_j)$, the value of ρ at the point (x_i, y_j)

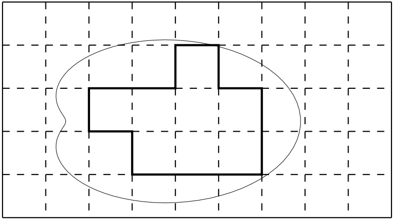


Then the mass m_{ij} of the rectangle is approximately

$$m_{ij} \approx \rho(x_i, y_j)\Delta A_{ij}$$

Step 3:

- Sum over all the rectangles lying *completely within A*.



This gives the first approximation to the total mass M

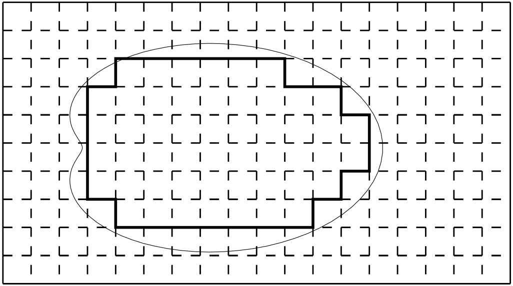
$$M \approx \sum_{\text{rectangles}} \rho(x_i, y_j)\Delta A_{ij}$$

that is

$$M \approx \sum_i \sum_j \rho(x_i, y_j)\Delta A_{ij}$$

Step 4:

- Reduce the size of the grid



- Repeat steps 2 and 3 with the new grid.

Since the grid provides a better covering of A and the approximation to $\rho(x, y)$ is better we obtain a better approximation to M.

Step 5:

- Make the grid size smaller and smaller
- then ΔA_{ij} becomes smaller and smaller
- and the approximation to $\rho(x, y)$ becomes better

Hence the approximation to M becomes better.

In the limit as $\Delta A_{ij} \to 0$ we obtain the true value of M

$$M = \lim_{\Delta A_{ij} \to 0} \left[\sum_i \sum_j \rho(x_i, y_j)\, \Delta A_{ij} \right].$$

If this limit exists we say that we have *integrated $\rho(x, y)$ over the region A*, written

$$\iint_A \rho(x, y) \, \mathrm{d}A.$$

This is called a *double integral*.

The area of one of the small rectangles is $\Delta A_{ij} = \Delta x_i \Delta y_j$, where Δx_i and Δy_i are the side lengths of the small rectangle in the x and y directions.

Therefore we can write

$$M = \iint_A \rho(x, y) \, \mathrm{d}A = \iint_A \rho(x, y) \, \mathrm{d}x\mathrm{d}y$$

Note

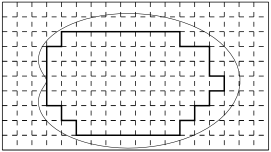
Many books write $\int_A$ instead of $\iint_A$, where it is understood that the one integral symbol actually stands for a double integral.

Other examples of how double integrals arise are the following.

- The total electrostatic charge on a region A carrying charge with density $\sigma(x,y)$ is
$$q = \iint_A \sigma(x,y)\,\mathrm{d}x\mathrm{d}y.$$
- The force exerted on a region A by a fluid exerting pressure $p(x,y)$ is
$$F = \iint_A p(x,y)\,\mathrm{d}x\mathrm{d}y.$$
- The volume under the surface $z = f(x,y)$ above the region A in the plane is
$$V = \iint_A f(x,y)\,\mathrm{d}x\mathrm{d}y.$$
- The area of the region A is
$$\iint_A \mathrm{d}x\mathrm{d}y.$$

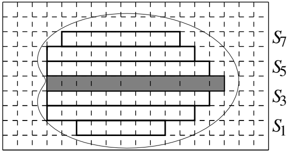
4.2 Evaluation of double integrals

We have defined what a double integral is, but how do we evaluate it? In step 3 we summed over all the rectangles



This can be carried out by

- Summing across each row forming the set of sums S_j



- Adding together these sums $S_1 + S_2 + \ldots = \sum_j S_j$.

Then we took the limit as $\Delta A_{ij} \to 0$. This is the same as taking the limits $\Delta x_i \to 0$ and $\Delta y_j \to 0$.

- Taking the limit as $\Delta x_i \to 0$ first has the result that each S_j is an integral with respect to x.
- then taking the limit as $\Delta y_j \to 0$ has the result that $\sum_j S_j$ is an integral with respect to y.

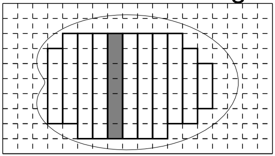
Therefore

$$M = \int \left[\int \rho(x, y)\,\mathrm{d}x \right] \mathrm{d}y.$$

This is called a *repeated integral* with respect to x first and then with respect to y.

Alternatively in step 3

- we could sum along each column forming the set of sums $\widehat{S}_i$,



- then adding together these sums $\widehat{S}_1 + \widehat{S}_2 + \cdots = \sum_i \widehat{S}_i$.
- On taking the limit $\Delta y_i \to 0$ first followed by the limit $\Delta x_i \to 0$ we are effectively carrying out integration with respect to y followed by integration with respect to x.
Therefore

$$M = \int \left[\int \rho(x, y) \, \mathrm{d}y \right] \mathrm{d}x,$$

a *repeated integral* with respect to y first and then x.

We can thus evaluate a *double integral* by converting it to a *repeated integral* either

- first with respect to x and then with respect to y

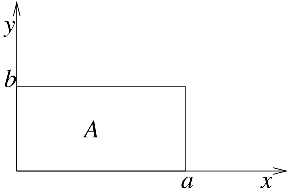
  or

- first with respect to y and then with respect to x.

4.3 Evaluation on a rectangular region

Example 4.3.1:

Integrate $f(x, y) = xy + 1$ over the region A defined
$0 \le x \le a,\ 0 \le y \le b$.



$$\begin{aligned}
\iint_A f(x,y)\,\mathrm{d}A &= \int_0^b \left[\int_0^a (xy+1)\ \mathrm{d}x\right]\ \mathrm{d}y \\
&= \int_0^b \left[\tfrac{1}{2}x^2y + x\right]_{x=0}^{x=a}\ \mathrm{d}y \\
&= \int_0^b \left(\tfrac{1}{2}a^2y + a\right)\ \mathrm{d}y \\
&= \left[\tfrac{1}{4}a^2y^2 + ay\right]_{y=0}^{y=b} \\
&= \tfrac{1}{4}a^2b^2 + ab
\end{aligned}$$

Changing the order of the integration

$$\begin{aligned}
\iint_A f(x,y)\,\mathrm{d}A &= \int_0^a \left[\int_0^b (xy+1)\ \mathrm{d}y\right]\ \mathrm{d}x \\
&= \int_0^a \left[\tfrac{1}{2}xy^2 + y\right]_{y=0}^{y=b}\ \mathrm{d}x \\
&= \int_0^a \left(\tfrac{1}{2}xb^2 + b\right)\ \mathrm{d}x \\
&= \left[\tfrac{1}{4}x^2b^2 + bx\right]_{x=0}^{x=a} \\
&= \tfrac{1}{4}a^2b^2 + ab
\end{aligned}$$

Note

It was necessary to find the limits of integration. In this example it was very easy because the region of integration was rectangular. **It will not always be so easy.**

4.4 Repeated integrals on a general region

Within a general region, determining the order of integration and the associated integration limits are key steps for a successful integration (if it can be done).

Some useful guidelines to determine the limits of integration are

1. Sketch the region of integration
2. Decide which variable to integrate first and draw a *typical* strip parallel to the direction of the first integration.

   (The end points of this strip give the limits of the first integral).
3. The limits between which this *typical* strip can lie within the region of integration give the limits of the second integral.

The general format for a repeated integral is of the form (i) or (ii):

(i)

$$\iint_A f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x,y)\,\mathrm{d}y \right) \mathrm{d}x$$

and the inner integral is evaluated first with integration over y and then the outer integral with x. Limits y_1 and y_2 depend on x in general and limits $x = a, b$ ensure cover of region A.

(ii)

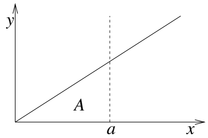
$$\iint_A f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_c^d \left(\int_{x_1(y)}^{x_2(y)} f(x,y)\,\mathrm{d}x \right) \mathrm{d}y$$

and the inner integral is evaluated first with integration over x and then the outer integral with y. Limits x_1 and x_2 depend on y in general and limits $y = c, d$ ensure cover of region A.

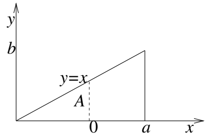
Example 4.4.1:

Evaluate $\iint_A xy \, \mathrm{d}A$, where A is the region between the curves $y = x$, $y = 0$ and $x = a$.

1. Sketch the region of integration.

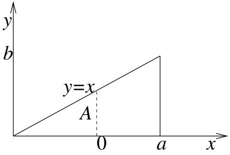


2. Integrate with respect to y first. So draw a typical strip parallel to the y axis.



The lower limit is $y = 0$, the upper limit is $y = x$.

3. The typical strip can lie between $x=0$ and $x=a$



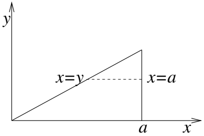
Hence

$$\iint_A xy \,\mathrm{d}A = \int_0^a \left[\int_0^x xy \,\mathrm{d}y\right] \mathrm{d}x.$$

$$\begin{aligned}
\int_0^a \left[\int_0^x xy \,\mathrm{d}y\right] \mathrm{d}x &= \int_0^a \left[\tfrac{1}{2}xy^2\right]_{y=0}^{y=x} \mathrm{d}x \\
&= \int_0^a \tfrac{1}{2}x^3 \,\mathrm{d}x \\
&= \left[\tfrac{1}{8}x^4\right]_{x=0}^{x=a} \\
&= \tfrac{1}{8}a^4
\end{aligned}$$

We can similarly carry out the integration with respect to x first and then with respect to y but the limits of integration are different.

1. Sketch the region of integration
2. Integrate with respect to x first, so draw a typical strip parallel to the x axis.



   The lower limit is $x = y$, the upper limit is $x = a$.
3. The typical strip can lie between $y = 0$ and $y = a$

Hence

$$\iint_A xy \, \mathrm{d}A = \int_0^a \left[\int_y^a xy \, \mathrm{d}x \right] \mathrm{d}y.$$

$$
\begin{aligned}
\int_0^a \left[\int_y^a xy \,\mathrm{d}x\right] \mathrm{d}y &= \int_0^a \left[\tfrac{1}{2}x^2 y\right]_{x=y}^{x=a} \mathrm{d}y \\
&= \int_0^a \left(\tfrac{1}{2}a^2 y - \tfrac{1}{2}y^3\right) \mathrm{d}y \\
&= \left[\tfrac{1}{4}a^2 y^2 - \tfrac{1}{8}y^4\right]_{y=0}^{y=a} \\
&= \tfrac{1}{4}a^4 - \tfrac{1}{8}a^4 \\
&= \tfrac{1}{8}a^4
\end{aligned}
$$

[Note: We do not need to use the bracket in the repeated integral. The notation is unambiguous without it.]

$$
\int_0^a \int_0^x xy \,\mathrm{d}y \,\mathrm{d}x = \int_0^a \int_y^a xy \,\mathrm{d}x \,\mathrm{d}y
$$

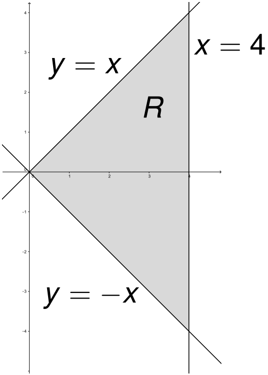
4.4.1 Reversing the order of integration

- Step 2 of calculating a double integral involves choosing which variable to integrate first.
- In some cases, the order of integration makes no difference to the calculation.
  - In some examples the calculation can be greatly simplified by reversing the order of integration.
  - Some integrals can *only* be computed if the order of integration is reversed.
- Note that reversing the order of integration does not change the value of the integral.

Example 4.4.2
Evaluate the integral

$$I = \iint_R x^2 + y^3 \, \mathrm{d}A,$$

where R is the triangular region bounded by $y = x$, $y = -x$ and $x = 4$.



Integrating with respect to x first,

$$I = \int_{-4}^{0} \int_{-y}^{4} x^2 + y^3 \, \mathrm{d}x \, \mathrm{d}y + \int_{0}^{4} \int_{y}^{4} x^2 + y^3 \, \mathrm{d}x \, \mathrm{d}y.$$

However, the calculation is made much simpler by reversing the order of integration:

$$I = \int_0^4 \int_{-x}^{x} x^2 + y^3 \, \mathrm{d}y \, \mathrm{d}x.$$

Evaluating the integral,

$$
\begin{aligned}
I &= \int_0^4 \left[x^2 y + \frac{1}{4} y^4 \right]_{y=-x}^{y=x} \mathrm{d}x, \\
&= \int_0^4 \left(x^3 + \frac{1}{4} x^4 \right) - \left(-x^3 + \frac{1}{4} x^4 \right) \mathrm{d}x, \\
&= \int_0^4 2x^3 \, \mathrm{d}x, \\
&= \left[\frac{1}{2} x^4 \right]_{x=0}^{x=4} = 128.
\end{aligned}
$$

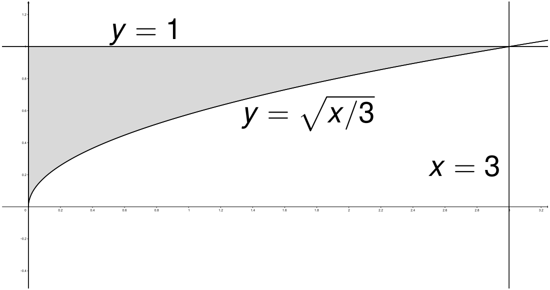
Exercise
Check that

$$
I = \int_{-4}^0 \int_{-y}^4 x^2 + y^3 \, \mathrm{d}x \, \mathrm{d}y + \int_0^4 \int_y^4 x^2 + y^3 \, \mathrm{d}x \, \mathrm{d}y = 128.
$$

Example 4.4.3
Evaluate the integral

$$I = \int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} \, \mathrm{d}y \, \mathrm{d}x.$$

Solution
It is impossible to integrate e^{y^3} with respect to y first, so we reverse the order of integration. To find the new limits, we sketch the integration region R.



Using horizontal strips instead, the integral becomes

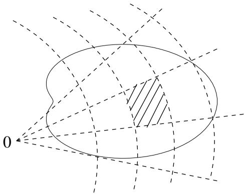
$$I = \int_0^1 \int_0^{3y^2} e^{y^3}\, \mathrm{d}x\, \mathrm{d}y.$$

It is now possible to proceed with the calculation.

$$\begin{aligned} I &= \int_0^1 \left[x e^{y^3} \right]_{x=0}^{x=3y^2} \mathrm{d}y, \\ &= \int_0^1 3y^2 e^{y^3}\, \mathrm{d}y, \\ &= \left[e^{y^3} \right]_{y=0}^{y=1}, \\ &= e - 1. \end{aligned}$$

4.5 Double integrals in polar coordinates

Initially we used a rectangular grid, with $\mathrm{d}A = \mathrm{d}x\mathrm{d}y$. We could have used a polar grid



In this case the area (shaded) element $\mathrm{d}A$ is given by

$$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta,$$

and

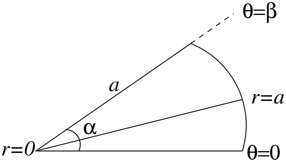
$$\iint_A f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint_{A'} F(r,\theta) r\,\mathrm{d}r\,\mathrm{d}\theta,$$

where $F(r,\theta) = f(x,y)$ with $x = r\cos\theta$, $y = r\sin\theta$, and A' is the region A described in polar coordinates.

Example 4.5.1

Find the area of the sector of a circle $0 \leq r \leq a$, $0 \leq \theta \leq \beta$.

- The required area is $\iint_A \mathrm{d}A = \iint_{A'} r \,\mathrm{d}r \,\mathrm{d}\theta$.



- $\iint_{A'} r \,\mathrm{d}r \,\mathrm{d}\theta = \int_0^\beta \int_0^a r \,\mathrm{d}r \,\mathrm{d}\theta = \int_0^\beta \left[\frac{1}{2} r^2\right]_0^a \mathrm{d}\theta$

  $= \int_0^\beta \frac{1}{2} a^2 \,\mathrm{d}\theta = \left[\frac{1}{2} a^2 \theta\right]_0^\beta = \frac{1}{2} a^2 \beta$.

This is reassuring: When $\beta = 2\pi$, the region is a circular disk, and the area is πa^2.

Example 4.5.2

If $f(x, y) = (25 - x^2 - y^2)$, A is the quadrant of the circle $x \geq 0$, $y \geq 0$, $x^2 + y^2 \leq 25$, and

$$J = \iint_A f(x, y)\,\mathrm{d}A,$$

(a) evaluate J in Cartesian coordinates.

(b) evaluate J by transforming to polar coordinates.

Solution

(a)

$$J = \int_0^5 \int_0^{(25-x^2)^{1/2}} (25 - x^2 - y^2)\,\mathrm{d}y\,\mathrm{d}x.$$

This is difficult and tedious to evaluate, so we will not bother!

(b)

$$J = \int_0^{\pi/2} \int_0^5 (25 - r^2) r \,\mathrm{d}r \,\mathrm{d}\theta.$$

This is much easier to evaluate:

$$\begin{aligned} J &= \int_0^{\pi/2} \left[\frac{25}{2} r^2 - \frac{1}{4} r^4 \right]_{r=0}^{r=5} \mathrm{d}\theta, \\ &= \int_0^{\pi/2} \frac{625}{4} \,\mathrm{d}\theta, \\ &= \left[\frac{625}{4} \theta \right]_{\theta=0}^{\theta=\pi/2}, \\ &= \frac{625\pi}{8}. \end{aligned}$$